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Estimation of Skeletal Muscle Mass and Visceral Adipose Tissue Volume by a Single Magnetic Resonance Imaging Slice in Healthy Elderly Adults[1,2]

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Abstract

Background: Assessing skeletal muscle (SM) and visceral adipose tissue (VAT) by a single MRI slice at lumbar vertebra (L) 3 can replace whole-body MRI in young and middle-aged adults. However, this technique has not been proven in older adults.
Objective: The aim of this analysis was to reinvestigate the best estimate for SM and VAT in an independent population of healthy elderly people.
Methods: SM and VAT were assessed by whole-body MRI in 84 subjects ≥60 y [45 men; mean ± SD age: 68.4 ± 5.4 y, mean ± SD body mass index (in kg/m^2): 25.5 ± 3.5]. SM and VAT areas of 9 slices at the lumbar spine were analyzed. The best estimate was investigated by Pearson correlations. Total volumes (in liters) were predicted by the area at lumbar vertebra 3 (A_{L3}). Besides Bland-Altman analysis, linear regressions were performed to explain the variance of the bias by age, height, and percentage of fat mass (%FM). In a mixed population (healthy elderly plus reference population), linear regression with total SM and VAT volume as dependent variables and A_{L3}, age, and height as independent variables was applied.
Results: When comparing the correlation coefficients between the tissue areas and total volumes, L3 was identified as the best estimate (*r* range: 0.71–0.94; all $P < 0.05$). However, Bland-Altman analysis showed a positive SM bias in men (mean ± SD: –1.0% ± 9.0%; $P < 0.05$) and a negative SM bias in women (mean ± SD: 3.7% ± 9.6%; $P < 0.05$). Contrary to SM, no significant bias was observed for VAT. In the elderly, stepwise linear regression showed height as a predictor for SM bias (R^2 = 0.21, SEE = 2.07 L; $P < 0.05$) and %FM and age as predictors of the nonsignificant VAT bias (R^2 = 0.26, SEE = 0.22L, $P < 0.05$), in men only. In the mixed population, A_{L3} and height were predictors for total SM, and A_{L3} for total VAT, independent of sex.
Conclusions: A_{L3} was confirmed as the best estimate for SM and VAT volumes in healthy elderly adults. Contrary to VAT, there is a bias for SM, and height has to be added to the algorithm. *J Nutr* 2016;146:2143–8.

Keywords: assessment of body composition, age-related changes in body composition, sarcopenia, sarcopenic obesity, advanced age, imaging techniques, skeletal muscle, visceral adipose tissue, simplification of the MRI protocol

Introduction

The assessment of skeletal muscle mass (SM)[6] and visceral adipose tissue (VAT) is used to identify sarcopenia and sarcopenic obesity. Aging and chronic diseases are associated with changes in body composition, i.e., the progressive loss of SM and gain in adipose tissue (particularly VAT) (1–4). To assess SM, imaging techniques, including whole-body MRI and computed tomography (CT), are gold standard methods (5, 6). However, these methods have some disadvantages, including radiation exposure with the use of CT, high costs, limited availability, and the

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[6] Abbreviations used: A_{L3}, area at lumbar vertebra 3; CT, computed tomography; L, lumbar vertebra; SAT, subcutaneous adipose tissue; SM, skeletal muscle; SM_{m-p}, measured minus predicted skeletal muscle; VAT, visceral adipose tissue; VAT_{m-p}, measured minus predicted visceral adipose tissue; %FM, percentage of fat mass.

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amount of time consumed by analyses. To simplify the MRI (and CT) protocol, several researchers have already used single slice areas to estimate whole-body composition. With the use of MRI, Shen et al. (7) found the best agreement with whole-body VAT at the exact positions 10 cm and 5 cm above the intervertebral disc between lumbar vertebra (L) 4 and L5 in men and women, which is in line with So et al. (8). Demerath et al. (9) proposed the area at lumbar vertebra 3 (A_{L3}) to be the best estimate for VAT in blacks and whites of both sexes. Other authors have found that the slice 5 cm above L4–L5 and slices at the thigh are good estimates for SM (10, 11). In a previous publication, we systematically investigated different slices at the lumbar spine, as well as at the thigh, in comparison with whole-body MRI scans and found L3 to be the best single reference slice for the estimation of SM, VAT, and subcutaneous adipose tissue (SAT) in healthy middle-aged adults (12). However, it remains to be proven whether L3 is a good estimate in older adults as well, and if these equations can be used for this specific age group. The objective of this study was to reinvestigate the best estimate for SM and VAT in an independent population of healthy elderly people. In addition, we wanted to validate the equations generated in middle-aged adults in the study population of elderly.

Methods

Subjects. The study sample included 84 healthy community-dwelling subjects (53% men) aged ≥60 y (60–81 y of age) with a BMI range (in kg/m^2) of 18.5–37.2. Exclusion criteria for all investigations were smoking, metallic implants, chronic or acute diseases, and medication, which influences body composition. The recruitment was conducted with the use of notice board postings, as well as local advertisement. Every participant completed examinations of body composition, strength, and mobility, which took place at the Institute of Human Nutrition (Christian-Albrechts-University, Germany). MRIs were performed at the Clinic for Diagnostic Radiology (University Medical Center Schleswig-Holstein, Kiel, Germany). All participants gave their written consent after we informed them about the purposes and methods of the investigation. The ethical committee of the University Kiel approved the study protocol.

In addition, we analyzed an age-mixed population (50% men, median age 49 y, and median BMI 25.3) by combining this elderly study population with the reference population from a previous work (13).

Anthropometric measurements. Body weight was measured to the nearest 0.01 kg with subjects wearing underwear by using an electronic scale coupled to a BOD POD (Life Measurement). Height was measured without shoes with a Seca stadiometer to the nearest 0.5 cm.

Measurement and prediction of body composition. The body composition of all subjects was measured by using whole-body MRI. Protocols were described in more detail elsewhere (13, 14). Tissues were obtained by using 1.5T scanners (Siemens Healthcare) with a T1-weighted gradient echo sequence. One-half of the elderly subjects were scanned with a MAGNETOM Avanto (repetition time 157 ms, echo time 15 ms, slice thickness 8 mm, and slice gap 2 mm). The remaining subjects were scanned with a MAGNETOM Vision (repetition time 575 ms, echo time 15 ms, slice thickness 10 mm, and gap 10 mm). All participants were assessed with the use of the same procedure previously described in a study population of middle-aged men and women (12). L3 was evaluated as a good estimate for the assessment of SM and VAT simultaneously. Based on this, sex-specific equations were established in a sample of 142 healthy subjects (67 men and 75 women). Whole-body SM was calculated by the formula $[V_{SM}(L)] = 0.141 \times A_{L3}\ (cm^2) + 3.790$ in women and volume of the skeletal muscle in liters $[V_{SM}(L)] = 0.136 \times A_{L3}\ (cm^2) + 5.944$ in men. Whole-body VAT was calculated as volume of the visceral adipose tissue in liters $[V_{VAT}(L)] = 0.026 \times A_{L3}\ (cm^2) + 0.121$ in women and $V_{VAT}[L] = 0.025 \times A_{L3}\ (cm^2) + 0.164$ in men (12).

Statistical analysis. Descriptive statistics were calculated and are presented as medians (IQRs). Differences between sexes were detected by the Mann-Whitney *U* test. A Wilcoxon test was used to test for significant differences between measured and predicted SM or VAT. Pearson correlations were performed to evaluate the strength of associations between whole-body volumes and areas at the lumbar spine (L1–L5); significant differences between the correlation coefficients were analyzed with the use of the method from Steiger (15). Bland-Altman plots were carried out to compare the tissues measured by the gold-standard whole-body MRI and predicted by the equations derived from single areas of SM and VAT at L3 (16). Pearson correlations were performed to evaluate the associations of the differences between measured minus predicted skeletal muscle (SM_{m-p}) and measured minus predicted visceral adipose tissue (VAT_{m-p}), with age, height, and percentage of fat mass (%FM). To determine the explained variance of SM_{m-p} and VAT_{m-p} from the independent variables age, height, and %FM, multiple linear regression models were performed with the use of SM_{m-p} and VAT_{m-p} as dependent variables.

In the age-mixed population, stepwise linear regression was done with total SM and VAT volumes as dependent variables and A_{L3}, age, and height as independent variables. All statistical analyses were done with SPSS, version 22.0). Two-sided tests of significance ($P < 0.05$) were used.

Results

The subjects' characteristics are summarized in **Table 1**. Men were taller and heavier and had greater BMI, SM, and VAT than women. Comparing the older adults with the reference population from our previous work (12), the elderly were smaller and had a lower weight and SM volume but a higher VAT volume (Mann-Whitney *U* test, $P < 0.05$) (Tables 1 and **2**).

With respect to investigating the best estimate for the whole-body tissues along the lumbar spine (L1–L5) in the elderly, all associations between areas and total volumes were significant. In both sexes, they were higher for VAT than they were for SM [method from Steiger (15)] (**Table 3**). Investigating L1–L5, the A_{L3} slice showed high or even the highest correlation coefficients with whole-body SM and VAT volume (Table 3), and it therefore was confirmed as the best estimate for SM and VAT in older adults. Accordingly, the equation with A_{L3} as the predictor was applied, which contributed to significantly higher values for the predicted SM than for the measured SM in women (SM_{m-p}: 4.0% ± 10.0%). By contrast, in men, no difference was observed (SM_{m-p}: 1.0% ± 9.1%). Independent of sex, predicted VAT volume was significantly lower than was measured VAT volume (VAT_{m-p} in men: −6.7% ± 17.4%; VAT_{m-p} in women: −8.3% ± 22.2%).

A comparison of whole-body MRI and the prediction from L3 is shown in **Figure 1** for SM and in **Figure 2** for VAT. The Bland-Altman analysis showed a significant bias of SM in men and women, even if SM_{m-p} was close to zero (men: −0.3 L, women: −0.7 L; both $P < 0.05$) (Figure 1). In both sexes, a moderate 95% limit of agreement was observed, which ranged from −4.9 to 4.3 L (−19.3% to 17.2%) in men and from −3.9 to 2.6 L (−23.9% to 15.9%) in women. A high SM was associated with an overestimation of the SM by L3 in women ($r = -0.34$, $P = 0.034$), but with an underestimation in men ($r = 0.34$, $P = 0.025$). Based on a linear regression model, a 1-L higher SM mean of the methods was associated with a 0.3-L decrease in SM_{m-p} in women and a 0.3-L increase in SM_{m-p} in men (both $P < 0.05$).

A calculation of Pearson correlation coefficients between the SM bias and age, height, and %FM showed associations with height in men only ($r = 0.46$, $P < 0.05$; for age see Figure 1). Stepwise linear regression analyses with SM_{m-p} as a dependent

TABLE 1 Characteristics of the older adults in the study[1]

	Total (n = 84)	Women (n = 39)	Men (n = 45)
Age, y	68.0 (64.0–70.8)	68.0 (63.0–73.0)	68.0 (65.0–70.0)
Height, m	1.71 (1.63–1.76)	1.63 (1.59–1.67)	1.75 (1.71–1.79)*
Weight, kg	72.6 (64.9–81.6)	65.0 (60.3–71.3)	79.6 (73.4–88.0)*
BMI, kg/m^2	25.3 (23.6–27.7)	24.1 (22.6–26.6)	25.9 (24.1–28.3)*
≤18.5	1.2	2.6	0.0
18.5–24.9	44.0	53.8	35.6
25–29.9	46.4	38.5	53.3
≥30	8.3	5.1	11.1
%FM	28.2 (22.0–34.4)	34.0 (27.7–40.0)	24.2 (21.4–28.6)*
MRI-assessed body composition, L			
SM	20.0 (16.7–25.5)	16.7 (14.9–17.8)	24.9 (23.1–28.3)*
SM_p	21.9 (17.4–25.7)	17.3 (15.7–18.4)#	25.5 (23.5–26.9)*
SM_{m-p}	−0.51 (−1.61 to 0.79)	−0.70 (−1.68 to 0.62)	−0.35 (−1.46 to 0.89)
TAT	22.1 (18.1–29.0)	25.6 (18.2–30.8)	20.5 (16.4–24.6)
VAT	3.0 (1.8–5.0)	2.1 (1.5–2.9)	4.2 (3.1–5.7)*
VAT_p	2.8 (1.6–4.7)	1.7 (1.1–2.7)#	4.5 (2.6–5.3)*#
VAT_{m-p}	0.1 (−0.1 to 0.8)	0.1 (−0.1 to 0.3)	0.1 (−0.3 to 1.0)
Single slice at L3			
SM area, cm^2	120.2 (95.6–145.2)	95.5 (84.7–103.5)	144.1 (128.9–154.4)*
VAT area, cm^2	102.5 (56.6–182.4)	62.5 (38.6–99.6)	172.8 (99.4–204.6)*

[1] Values are medians (IQRs) or percentages. *Significantly different from women, $P < 0.05$ (Mann-Whitney U test). #Significantly different from measured values, $P < 0.05$ (Wilcoxon test). L3, lumbar vertebra 3; SM, skeletal muscle; SM_{m-p}, measured minus predicted skeletal muscle; SM_p, predicted skeletal muscle; TAT, total adipose tissue; VAT, visceral adipose tissue; VAT_{m-p}, measured minus predicted visceral adipose tissue; VAT_p, predicted visceral adipose tissue; %FM, percentage of fat mass.

variable and age, height, and %FM as independent variables found height as the only predictor of SM_{m-p} in men, with an explained variance of 21% (SEE = 2.07 L; $P < 0.05$).

When compared with SM, there was no systematic error in the prediction of VAT by L3 (men: VAT_{m-p} = 0.3 L; women: within 95% of the individual VAT_{m-p} = 0.2 L; both $P > 0.05$) (see Figure 2). The limits of agreement, within 95% of the individual VAT_{m-p} values of the study population, ranged from −28.0% to 41.5% (−1.3 to 1.9 L) in men and from −36.4% to 52.6% (−0.8 to 1.2 L) in women. Pearson correlations between VAT_{m-p} and age, height, and %FM showed significant relations for age and %FM in men only [age: $r = -0.40$ (Figure 2); %FM: $r = 0.42$; both $P < 0.05$). Accordingly, whole-body VAT volume tended to be underestimated by L3 in younger elderly men, as well as in men with a higher %FM. By contrast, an overestimation was seen in older elderly men and at a lower %FM.

TABLE 2 Characteristics of the mixed and reference populations[1]

	Mixed population[2]		Reference population	
	Women (n = 114)	Men (n = 112)	Women (n = 75)	Men (n = 67)
Age, y	48.0 (32.0–64.3)	49.5 (37.0–66.8)	35.0 (29.0–48.0)#	39.0 (27.0–46.0)#
Height, m	1.67 (1.62–1.72)*	1.78 (1.73–1.82)	1.69 (1.64–1.73)*,#	1.81 (1.76–1.85)#
Weight, kg	68.6 (61.5–78.2)*	83.4 (74.9–95.1)	69.3 (61.6–87.5)*,#	86.3 (77.6–98.5)#
BMI, kg/m^2	24.3 (22.1–28.3)*	26.1 (23.7–28.9)	24.4 (22.1–31.1)	26.3 (23.2–30.0)
≤18.5	0.9	0.0	0.0	0.0
18.5–24.9	53.5	37.5	53.3	38.8
25–29.9	25.4	42.9	18.7	35.8
≥30	20.2	19.6	28.0	25.4
%FM	32.1 (26.9–40.0)*	22.7 (18.2–27.1)	21.7 (16.9–26.2)*	30.7 (26.8–40.6)#
MRI-assessed body composition, L				
SM	19.0 (16.9–21.6)*	28.2 (25.5–32.5)	20.7 (18.6–22.9)*,#	31.9 (27.6–34.2)#
TAT	23.5 (18.2–31.8)*	20.2 (15.6–25.5)	22.3 (18.4–37.9)*	20.0 (13.4–26.7)
VAT	1.4 (0.6–2.5)*	3.2 (1.6–4.9)	0.9 (0.4–2.2)*,#	2.3 (1.1–3.8)#
Single slice at L3				
SM area, cm^2	113.2 (97.2–128.5)*	168.5 (145.4–191.3)	122.4 (108.9–133.9)*,#	183.6 (170.0–199.4)#
VAT area, cm^2	46.0 (15.5–87.1)*	120.6 (67.2–180.2)	35.0 (10.2–75.5)*,#	91.7 (42.1–149.6)#

[1] Values are medians (IQRs) or percentages. *Significantly different from men of the same population. #Significantly different from the older adults of the same sex, $P < 0.05$ (Mann-Whitney U test). L3, lumbar vertebra 3; SM, skeletal muscle; TAT, total adipose tissue; VAT, visceral adipose tissue; %FM, percentage of fat mass.

[2] Includes reference population and older adults.

TABLE 3 Pearson correlations between total tissue volumes of VAT, SM, and single-slice areas in older adults[1]

	Correlation coefficients								
	L1	L1–L2	L2	L2–L3	L3	L3–L4	L4	L4–L5	L5
Women									
(n = 39)									
SM	0.34[c,d,e]	0.47[c,e]	0.60[c,e]	0.65[e]	0.71[e]	0.69[e]	0.68[e]	0.53[b,c,e]	0.37[b,c,e]
VAT	0.63[c]	0.90[c]	0.95	0.94	0.94	0.92	0.89[c]	0.88[b,c]	0.89[b,c]
Men									
(n = 45)									
SM	0.67	0.69[e]	0.71	0.80[e]	0.76[e]	0.77[e]	0.70[e]	0.68[e]	0.64[e]
VAT	0.59[c]	0.83[c]	0.56[c]	0.92	0.93	0.91	0.91	0.88	0.88[c]

[1] All correlation coefficients are significant at $P < 0.01$. [b] n = 38 subjects; [c] significantly different from correlation coefficients at L3 (within tissue and sex); [d] significantly different from correlation coefficients in men; [e] significantly different from correlation coefficients of VAT; $P < 0.05$, method from Steiger (14). L, lumbar vertebra; SM, skeletal muscle; VAT, visceral adipose tissue.

In a stepwise linear regression analysis in men, with VAT_{m-p} as the dependent variable and age, height, and %FM as independent variables, %FM was entered into the equation first (R^2 = 0.18; SEE = 0.7 L), followed by age (R^2 change = 0.08; SEE change = 0.03 L). On the basis of the multivariate regression model, a 1% higher fat mass and a 1-y younger age were associated with a 0.05-L increase in VAT_{m-p} and, therefore, with a higher underestimation of VAT by L3 (P < 0.05).

Stepwise linear regression models were considered in an age-mixed population that included the reference population and the older adults. Total SM and VAT volume represented the dependent variables, and A_{L3}, age, and height represented the independent variables (**Table 4**). For SM, A_{L3} was entered into the regression equation first, independent of sex, followed by height (P < 0.05). For VAT and both sexes, only A_{L3} was included in the analysis, explaining a major part of the variance in women and men (P < 0.05).

Discussion

In a previous investigation on a reference population of middle-aged adults, we found that the A_{L3} single slice was the best tradeoff to predict whole-body SM, SAT, and VAT (12). An algorithm to predict whole-body SM and VAT volume was generated (11). However, that study was limited to a population with an age range of 19–65 y. Previously, other authors investigated single slices as predictors for total tissue volumes (7–9, 11), but they did not distinguish between different age groups or they examined middle-aged subjects only. We hypothesized that advanced age would affect accuracy in predicting SM and VAT volumes because of age-related changes in body composition.

To our knowledge, the present study is the first to investigate the precision of predicting whole-body SM and VAT volume by a single MRI slice in the elderly. We were able to confirm that the A_{L3} is the best estimate for total SM and VAT in healthy elderly (Table 3). Measured and predicted SM and VAT were compared with the use of the gold-standard whole-body MRI and the algorithm based on the reference population. However, we found a significant bias of SM (positive for men and negative for women), with moderate limits of agreement (Figure 1). Particularly in men, the positive SM bias would overestimate SM in persons with a low SM, and the formula thus would mask sarcopenia. Differences in anthropometric data (especially height) between the study samples of middle-aged and elderly subjects might be responsible in part for the SM bias. Some researchers have found a reduction in height caused by age as a result of, e.g., decreased bone density (16–20), which could result in a diminished prediction power for A_{L3}. In accordance with

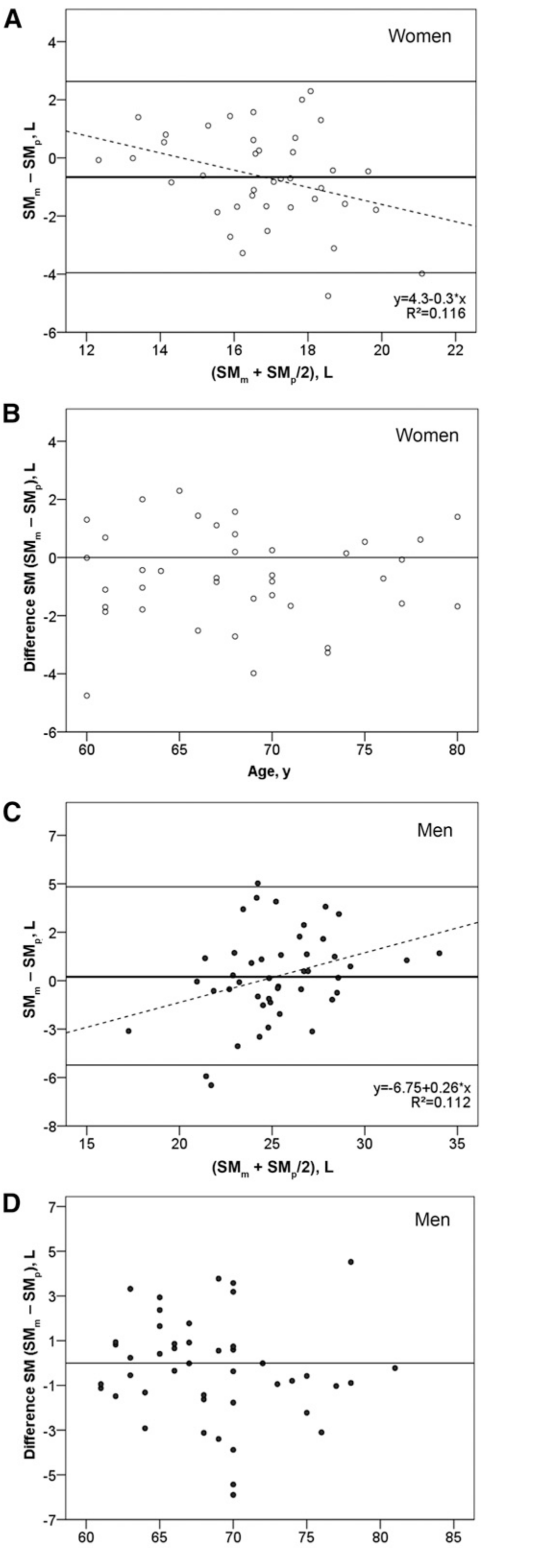

FIGURE 1 Bland-Altman plots of the differences between SM_m and SM_p (A and C) and the association of the mean differences with age (B and D) in older men and women. The bold line is the mean difference, and the horizontal lines are mean differences ± 2 SD. SM, skeletal muscle; SM_m, measured skeletal muscle; SM_p, predicted skeletal muscle.

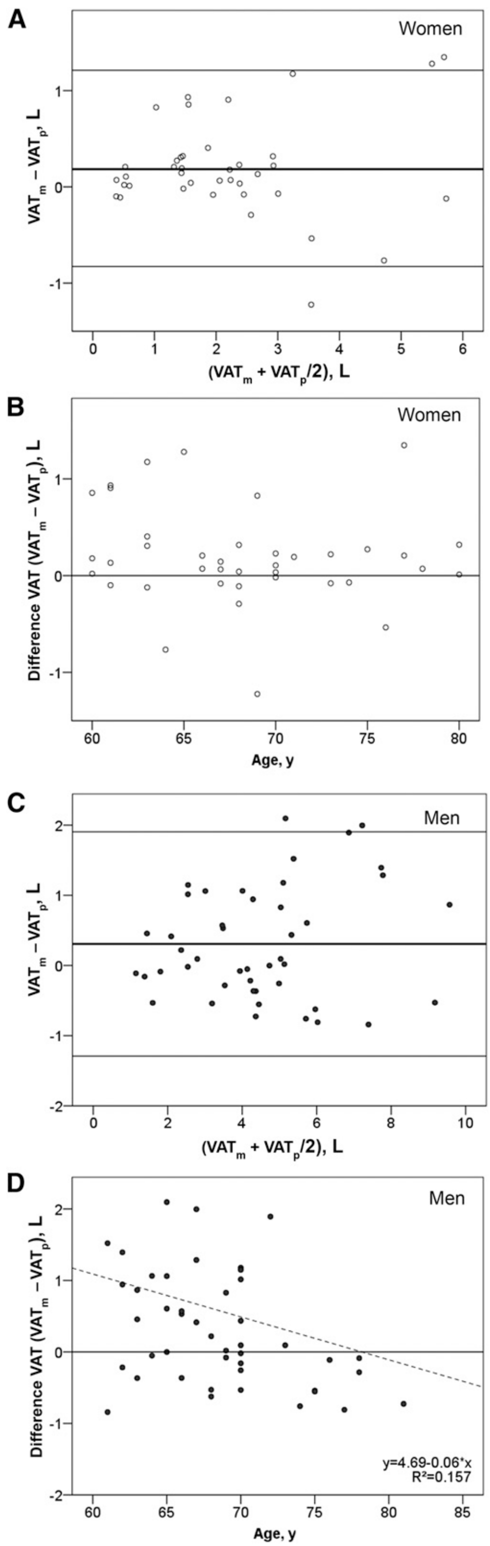

FIGURE 2 Bland-Altman plots of the differences between VAT_m and VAT_p (A and C) and the association of the mean differences with age (B and D) in older men and women. The bold line is the mean difference, and the horizontal lines are mean differences ± 2 SD. VAT_m, measured visceral adipose tissue; VAT_p, predicted visceral adipose tissue.

such anatomical changes, 21% of the SM bias variance was explained by height in elderly men, which was confirmed by an increased variance explanation of total SM by the height in the regression analyses of the mixed-population model (Table 4).

To date, to our knowledge, the majority of studies have investigated the estimation of SM by single-slice areas in a cross-sectional setting (11, 22); thus, little is known about SM changes of specific cross-sectional areas. Yet we already know that the progressive loss of muscle mass that accompanies aging differs between body regions. For instance, there is a predominant SM depletion at the lower limbs resulting in a high correlation between the thigh area and whole-body volume of SM (2, 3); thus, L3 may not accurately reflect SM changes (12). However, in contrast with that of L3, the area at the midthigh contains SM and SAT only; thus, L3 is required to determine the metabolic risk associated with VAT.

With respect to VAT, no bias of the VAT volume was observed when comparing the gold standard and prediction by L3 (Figure 2). VAT_{m-p} was negatively associated with age and positively associated with %FM, although only in men. Thus, the lower the age, the higher the underestimation of VAT by L3, which could be caused by the negative relation between the total VAT volume and age that we found in men. On the contrary, in middle-aged men, we found a positive association between VAT volume and age. This is in line with previous investigations that found an increased VAT volume with age (23–26). In addition, a study by Jackson et al. (27) in 7265 men aged 20–96 y showed increased fat mass, which was followed by a slight decrease in the eighth decade of life. In contrast with this finding, Roubenoff et al. (28) found a peak of fat mass in the fifth or sixth decade of life, which is in agreement with our results. However, most study designs did not distinguish between different regional fat masses. We assumed that the association between %FM and VAT_{m-p} was due to the negative relation between age and %FM; however, this was not significant in our study sample.

In women, age is related to a shift from peripheral to central and more androgenic fat distribution, which is well described by several authors (29–32). However, contrary to men, we found no associations of VAT_{m-p} with age, height, or %FM in women. Several studies have shown that the increase in VAT is associated with the menopausal transition and accompanying endocrine changes, in addition to the aging process itself (29, 30, 33–35). Janssen et al. (34) observed an association between VAT and plasma testosterone concentrations that was independent of age, race, %FM, or other cardiovascular disease risk factors in 359 women. The individual age of menopause varies between several decades of age, and this may be a reason for the missing

TABLE 4 Association of total tissue volumes of VAT and SM with age, height, and A_{L3} slice in the mixed population[1]

Variables	B coefficient	Intercept	R^2	SEE	P
Women (n = 114)					
SM					
A_{L3}	0.138	3.910	0.755	1.87	<0.001
A_{L3} + height	17.715	−22.885	0.836	1.54	<0.001
VAT					
A_{L3}	0.026	0.186	0.937	0.36	<0.001
Men (n = 112)					
SM					
A_{L3}	0.137	5.610	0.807	2.14	<0.001
A_{L3} + height	19.663	−26.877	0.857	1.85	<0.001
VAT					
A_{L3}	0.026	0.205	0.904	0.70	<0.001

[1] Model estimates are from stepwise multiple regressions with age, height, and A_{L3} slice as independent variables and total tissue volumes as dependent variables. A_{L3}, area at lumbar vertebra 3; SM, skeletal muscle; VAT, visceral adipose tissue.

association of VAT_{m-p} and age in women. For total VAT, the stepwise regression analysis in the mixed population with A_{L3}, age, and height as independent variables showed A_{L3} as the only predictor; hence, we assumed a good estimation of VAT volume when we used the algorithm in the elderly.

This was a cross-sectional investigation that was conducted in healthy Caucasians aged 60–81 y. Thus, no evidence is given for very elderly subjects (≥81 y of age) or ill patients. The presented results are not transferable to other ethnicities. Fat infiltrations of SM may also add to our analysis.

To summarize, our findings indicate that a single MRI slice at L3 is the best estimate for SM and VAT in healthy elderly subjects. For VAT, the algorithm published before therefore can be applied in older adults. For SM, there is a bias, and height has to be added in the formula to ensure a correct assessment of SM.

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